

Human Review: Square-zero unitization and non-open multiplication

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed example appears to correctly answer the first clause of Open Problem 3 in Dragă-Kania as printed. Namely, it gives a commutative unital Banach algebra whose maximal ideal space is zero-dimensional, but whose multiplication map is not open.

The construction is the unitization of a square-zero ideal. Let

$$A = \mathbb{C} \oplus E, \quad E = \mathbb{C}^2,$$

with multiplication

$$(\lambda, x)(\mu, y) = (\lambda\mu, \lambda y + \mu x)$$

and norm $\|(\lambda, x)\| = |\lambda| + \|x\|_\infty$. The algebraic and norm checks are straightforward: A is a commutative unital Banach algebra, and the norm is submultiplicative.

Maximal Ideal Space

The computation of the maximal ideal space is correct. If $\varphi : A \rightarrow \mathbb{C}$ is a character and $x \in E$, then

$$(0, x)^2 = 0.$$

Therefore

$$\varphi(0, x)^2 = \varphi((0, x)^2) = 0,$$

so $\varphi(0, x) = 0$. Since any character on a unital complex Banach algebra satisfies $\varphi(1) = 1$, it follows that

$$\varphi(\lambda, x) = \lambda.$$

Thus A has exactly one character, so $\text{Max}(A)$ is a singleton and hence zero-dimensional.

Failure of Openness

The failure of openness is also correct. Let $e_1 = (1, 0)$, $e_2 = (0, 1)$ in E , and put

$$a = (0, e_1).$$

Then $a^2 = 0$. To show that multiplication is not open at (a, a) , it is enough to find $0 < \varepsilon < 1$ such that

$$B_A(a, \varepsilon) \cdot B_A(a, \varepsilon)$$

is not a neighbourhood of 0.

Given any $\delta > 0$, choose $0 < t < \delta$ and consider

$$z = (0, te_2).$$

Then $\|z\| < \delta$. Suppose, for contradiction, that

$$z = (\alpha, x)(\beta, y)$$

with both factors in $B_A(a, \varepsilon)$. Comparing scalar coordinates gives

$$\alpha\beta = 0.$$

Hence $\alpha = 0$ or $\beta = 0$. If $\alpha = 0$, then the ideal coordinate equation becomes

$$\beta x = te_2.$$

Since $t \neq 0$, one has $\beta \neq 0$, and so $x \in \mathbb{C}e_2$. But every vector in $\mathbb{C}e_2$ is at distance at least 1 from e_1 in the supremum norm:

$$\|x - e_1\|_\infty \geq 1.$$

This contradicts the assumption that

$$(\alpha, x) \in B_A(a, \varepsilon), \quad \varepsilon < 1.$$

The case $\beta = 0$ is identical, using y instead of x .

Thus every ball around $0 = a^2$ contains points $(0, te_2)$ which are not in the product of the two ε -balls around a . Therefore multiplication is not open.

Review Outcome

The proof appears correct. It is a simple and effective counterexample to the first clause of the open problem as literally stated.

The main caveat is semantic rather than mathematical: the example is highly non-semisimple, and it has only the trivial idempotents. Thus it does not answer the stronger dense-idempotent follow-up, and it may be worth asking the authors whether they intended an additional assumption such as semisimplicity. As written, however, the counterexample is valid.